chloroplasts	START: Roughly, what is the maximum magnification of a school microscope.	up to x400	What do school microscopes use to reveal a specimen on a slide?
light	How many times can an electron microscope magnify an object?	x2 000 000	Your eyepiece lens is x3 and the objective lens is x75. What is the magnification?
x225	How do you work out the actual size of a specimen if you know the image size and the magnification?	image ÷ magnification	How many micrometres in 1 millimetre?
1 000	How many millimetres in 1 centimetre?	10	How many millimetres is 500 micrometres (µm)?
0.5	The power of a microscope to show that two points are separate is known as..?	resolving power	How large is the image if you magnify a 3 mm long head louse by x5?
15 mm	Which type of cells are animal and plant cells?	eukaryotic cells	Bacteria do not have a nucleus. What type of organism are they?
prokaryotes	Prokaryotic cells contain rings of DNA which are called..?	plasmids	A molecule is 2 nm long; a virus 200 nm long. How many orders of magnitude larger is the virus?
10^2	Some bacteria have a flagellum. What's its purpose?	movement	Which type of specialised cell has dendrites?

nerve cell (neurone)	Which type of cell has special proteins to make fibres contract?	striated muscle cell	What are suspended in cell cytoplasm?
organelles	Where does aerobic respiration take place in a cell?	in mitochondria	How long are mitochondria?
1 – 2 µm	Which single organelle in a cell has an average diameter of 10 µm?	nucleus	What structure prevents some substances from entering the cell?
the cell membrane	What surrounds the cell membrane of a plant cell?	a cell wall	What is the role of cellulose in the cell wall?
It strengthens and supports the cell wall	How long are chloroplasts?	3 – 5 µm	How many chloroplasts would fit along the diameter of the average cell nucleus?
2 or 3	Which organelles are responsible for protein synthesis?	ribosomes	Do plant cells have mitochondria?
yes!	Which part of a sperm cell stores a digestive enzyme?	the acrosome	Which plant cell, involved in water transport, is strengthened by spirals of lignin?
xylem	Which plant cells transport dissolved nutrients?	phloem	What structures contain chlorophyll?

Teaching notes

There are 32 cards, allowing it to be used with a whole class. The activity could also be done individually, in pairs or in small groups to produce a complete ordered set of cards.